MODULE V

COMPLEX VARIABLE – RESIDUE INTEGRATION

(Text 2: Relevant topics from 16.1, 16.2, 16.3, 16.4)

Syllabus : Laurent's series (without proof), zeros of analytic functions, singularities, poles, removable singularities, essential singularities, Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral using residue theorem, Residue integration of real integrals – integrals of rational functions of $\cos\theta$ and $\sin\theta$, integrals of improper integrals of the form $\int_{-\infty}^{\infty} f(x)\,dx$ with no poles on the real axis.

Laurent's Series

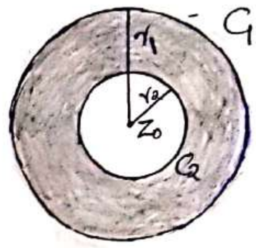
If we want to develop a function $f(z)$ in powers of $z-z_0$ when it is not analytic at z_0 we cannot use a Taylor series. We use a new kind of series, called Laurent's series consisting of positive integer powers of $z-z_0$ (and a constant) as well as negative integer powers of $z-z_0$.

Theorem (Laurent's Theorem)

Let $f(z)$ be analytic in a domain consisting two concentric circles C_1 and C_2 with centre z_0 and radii r_1 and r_2 $(r_1 > r_2)$ and the annulus between them. Then for any z for which $r_1 < |z-z_0| < r_2$, $f(z)$ can be represented by the Laurent's series

$$\boxed{\begin{aligned} f(z) &= a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots + \frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \cdots \\ &= \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n} \end{aligned}}$$

where $a_n = \frac{1}{2\pi i} \oint_{C_1} \frac{f(\omega)}{(\omega - a)^{n+1}} d\omega$; $n = 0, 1, 2, \cdots$



$b_n = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\omega)}{(\omega - a)^{-n+1}} d\omega$; $n = 1, 2, 3, \cdots$

integration being taken in counter clockwise around any simple closed path C that lies in the annulus and encircles the inner circle.

Note: i) $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ is called regular part of $f(z)$

and $\sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$ is called principal part of $f(z)$

ii) To find the Laurent's series, simply expand the function about required point using binomial series instead of finding a_n & b_n by complex integration.

iii) Laurent's series of a given analytic function in the annulus of convergence is unique. The function in two different annulus with same centre can have different Laurent's series.

Results: $(1+z)^{-n} = 1 - nz + \frac{1}{2!} n(n+1) z^2 - \frac{1}{3!} n(n+1)(n+2) z^3 + \cdots$ valid in $|z| < 1$

$(1-z)^{-n} = 1 + nz + \frac{1}{2!} n(n+1) z^2 + \frac{1}{3!} n(n+1)(n+2) z^3 + \cdots$ valid in $|z| < 1$

Problems:

1) Find the Laurent's series of $z^{-5} \sin z$ with centre 0

Ans: $f(z) = \frac{\sin z}{z^5}$ [$f(z)$ is not analytic at $z=0$]

$= \frac{1}{z^5} \left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots \right]$

$$= \frac{1}{z^4} - \frac{1}{3!z^2} + \frac{1}{5!} - \frac{z^2}{7!} + \cdots$$

$$= \frac{1}{z^4} - \frac{1}{6z^2} + \frac{1}{120} - \frac{1}{5040}z^2 + \cdots \quad [|z|>0]$$

[Here the annulus is the whole complex plane without the origin & the principal part of the series is $\frac{1}{z^4} - \frac{1}{6z^2}$]

2. Find the Laurent's series of $z^2 e^{1/z}$ with centre '0'

<u>Ans:</u> $f(z) = z^2 e^{1/z}$

$$= z^2\left[1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots\right]$$

$$= z^2 + z + \frac{1}{2} + \frac{1}{3!z} + \frac{1}{4!z^2} + \cdots \quad [|z|>0]$$

<u>Result</u> $\left[e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \right.$

$\left. e^{1/z} = 1 + \frac{1}{1!z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots \right]$ (z replaced by $1/z$)

Here the <u>principal part</u> is an infinite series.

3. Develop $\frac{1}{1-z}$ (i) in non-negative powers of z
(ii) in negative powers of z.

<u>Ans:</u> $f(z) = \frac{1}{1-z}$ is not analytic at $z=1$

$\therefore$ Regions are $|z|>1$ and $|z|<1$

$f(z) = \frac{1}{1-z}$ is analytic in the regions $|z|>1$ & $|z|<1$

(i) $f(z) = \frac{1}{1-z} = (1-z)^{-1} = 1 + z + z^2 + \cdots$

is valid in the region $|z|<1$

ii) $f(z) = \frac{1}{1-z} = \frac{1}{-z(1-\frac{1}{z})} = -\frac{1}{z}\left[1 - \frac{1}{z}\right]^{-1}$

$$= -\frac{1}{z}\left[1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \cdots\right]$$

$$= -\frac{1}{z} - \frac{1}{z^2} - \frac{1}{z^3} - \frac{1}{z^4} - \cdots \quad \text{valid in } \left|\frac{1}{z}\right| < 1 \Rightarrow |z| > 1$$

Results : $(1-z)^{-1} = 1 + z + z^2 + \cdots$ valid in $|z| < 1$

4. Find all Laurent's series of $\frac{1}{z^3 - z^4}$ with centre '0'

Ans: $f(z) = \frac{1}{z^3 - z^4}$

$f(z) = \frac{1}{z^3 - z^4}$
$= \frac{1}{z^3(1-z)}$
not analytic at $z=0$ & $z=1$
$\therefore f(z)$ is analytic in (i) $0 < |z| < 1$
(ii) $|z| > 1$

i) $$\frac{1}{z^3 - z^4} = \frac{1}{z^3(1-z)}$$
$$= \frac{1}{z^3}(1-z)^{-1}$$
$$= \frac{1}{z^3}(1 + z + z^2 + z^3 + z^4 + z^5 + \cdots)$$
$$= \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \cdots$$
valid in $0 < |z| < 1$

ii) $$\frac{1}{z^3 - z^4} = \frac{1}{-z^4\left(1 - \frac{1}{z}\right)}$$
$$= -\frac{1}{z^4}\left(1 - \frac{1}{z}\right)^{-1}$$ valid in $\left|\frac{1}{z}\right| < 1$
$$= -\frac{1}{z^4}\left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right)$$
valid in $|z| > 1$
$$= -\frac{1}{z^4} - \frac{1}{z^5} - \frac{1}{z^6} - \cdots$$

5. Expand $f(z) = \frac{z}{(z+1)(z+2)}$ as a Laurent's series about $z = -2$ in $0 < |z+2| < 1$

Ans : $f(z) = \frac{z}{(z+1)(z+2)}$

Laurent's series about $z = -2$
$\Rightarrow$ series in powers of $(z+2)$

By method of partial fractions

$$\frac{z}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2}$$
$$\Rightarrow \quad z = A(z+2) + B(z+1)$$

Put $z=-2 \Rightarrow -2 = -B \Rightarrow B=2$

Put $z=-1 \Rightarrow -1 = A \Rightarrow A=-1$

$$\therefore \frac{z}{(z+1)(z+2)} = \frac{-1}{z+1} + \frac{2}{z+2}$$

$$= \frac{2}{z+2} - \frac{1}{z+1}$$

$$= \frac{2}{z+2} - \frac{1}{(z+2)-2+1}$$

[Arranged suitably to make the binomial expansion valid in $0<|z+2|<1$]

$$= \frac{2}{z+2} - \frac{1}{(z+2)-1}$$

$$= \frac{2}{z+2} - \frac{1}{-[1-(z+2)]}$$

$$= \frac{2}{z+2} + [1-(z+2)]^{-1}$$

$$= \frac{2}{z+2} + [1+(z+2)+(z+2)^2+(z+2)^3+ \cdots]$$

provided $0<|z+2|<1$

6. Expand $\frac{z}{(z-1)(z-2)}$ in (i) $0<|z-2|<1$ (ii) $|z-1|>1$

Ans: $f(z) = \frac{z}{(z-1)(z-2)}$

By partial fraction method

$$\frac{z}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$\Rightarrow z = A(z-2) + B(z-1)$$

Put $z=2 \Rightarrow 2 = B \quad \therefore B=2$

Put $z=1 \Rightarrow 1 = -A \quad \therefore A=-1$

$$\therefore \frac{z}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{2}{z-2}$$

(i) in the region $0<|z-2|<1$

$$\frac{z}{(z-1)(z-2)} = \frac{-1}{(z-2)+2-1} + \frac{2}{z-2}$$

$$= \frac{-1}{(z-2)+1} + \frac{2}{z-2}$$

$$= -\frac{1}{1+(z-2)} + \frac{2}{z-2}$$

$$= -\left[1+(z-2)\right]^{-1} + \frac{2}{z-2}$$

$$= -\left[1-(z-2)+(z-2)^2-(z-2)^3+\cdots\right] + \frac{2}{z-2}$$

$$= -1+(z-2)-(z-2)^2+(z-2)^3-\cdots+\frac{2}{z-2}$$

valid in $|z-2|<1$

ii) $$\frac{z}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{2}{z-2}$$

$$= -\frac{1}{z-1} + \frac{2}{(z-1)+1-2}$$

$$= -\frac{1}{z-1} + \frac{2}{(z-1)-1}$$

$$= \frac{-1}{z-1} + \frac{2}{(z-1)\left[1-\frac{1}{z-1}\right]}$$

$$= \frac{-1}{z-1} + \frac{2}{z-1}\left[1-\frac{1}{z-1}\right]^{-1}$$

$$= \frac{-1}{z-1} + \frac{2}{z-1}\left[1+\frac{1}{z-1}+\left(\frac{1}{z-1}\right)^2+\cdots\right]$$

$$= \frac{-1}{z-1} + \frac{2}{z-1} + \frac{2}{(z-1)^2} + \frac{2}{(z-1)^3} + \cdots$$

valid in $|z-1|>1$

$$\boxed{|z-1|>1 \Rightarrow \left|\frac{1}{z-1}\right|<1}$$

(Arranged suitably to make the binomial expansion valid.)

7 Expand $\frac{e^{2z}}{(z-1)^3}$ about $z=1$ as a Laurent's series

Ans: $$f(z) = \frac{e^{2z}}{(z-1)^3}$$

$$= \frac{e^{2(z-1+1)}}{(z-1)^3}$$

$$= \frac{e^{2(z-1)}\cdot e^2}{(z-1)^3}$$

$$= \frac{e^2}{(z-1)^3}\left[e^{2(z-1)}\right]$$

Laurent's series about $z=1$
$\Rightarrow$ terms in powers of $z-1$

$$= \frac{e^2}{(z-1)^3}\left[1 + \frac{2(z-1)}{1!} + \frac{[2(z-1)]^2}{2!} + \frac{[2(z-1)]^3}{3!} + \cdots\right]$$

[using the expansion of $e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots$]

$$= \frac{e^2}{(z-1)^3}\left[1 + 2(z-1) + \frac{4(z-1)^2}{2!} + \frac{8(z-1)^3}{3!} + \cdots\right]$$

$$= e^2\left[\frac{1}{(z-1)^3} + \frac{2}{(z-1)^2} + \frac{2}{z-1} + \frac{4}{3} + \cdots\right]$$

8. Expand $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ in the region

i) $|z| < 2$ (ii) $2 < |z| < 3$ (iii) $|z| > 3$

Ans: $f(z) = \frac{z^2-1}{(z+2)(z+3)} = 1 + \frac{-5z-7}{(z+2)(z+3)}$ ——— (1) by actual division

$$z^2+5z+6 \overline{\left) \begin{array}{l} z^2-1 \\ \underline{z^2+5z+6} \\ \quad -5z-7 \end{array} \right.} \quad 1$$

By method of partial fraction,

$$\frac{-5z-7}{(z+2)(z+3)} = \frac{A}{z+2} + \frac{B}{z+3}$$

$\Rightarrow -5z-7 = A(z+3) + B(z+2)$

Put $z=-3$, $8 = -B \Rightarrow B = -8$

Put $z=-2$ $3 = A \Rightarrow A = 3$

(1) $\Rightarrow f(z) = 1 + \frac{3}{z+2} + \frac{-8}{z+3}$ ——— (2)

i) $|z| < 2 \Rightarrow \left|\frac{z}{2}\right| < 1$ & $\left|\frac{z}{3}\right| < 1$

$\therefore f(z) = 1 + \frac{3}{2(1+\frac{z}{2})} + \frac{-8}{3(1+\frac{z}{3})}$ [Arranged (2) suitably to make the binomial expansion valid]

$$= 1 + \frac{3}{2}\left(1+\frac{z}{2}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$$

$$= 1 + \frac{3}{2}\left[1 - \left(\frac{z}{2}\right) + \left(\frac{z}{2}\right)^2 - \left(\frac{z}{2}\right)^3 + \cdots\right] - \frac{8}{3}\left[1 - \left(\frac{z}{3}\right) + \left(\frac{z}{3}\right)^2 + \cdots\right]$$

$$= 1 + \frac{3}{2}\left[1 - \frac{z}{2} + \frac{z^2}{2^2} - \frac{z^3}{2^3} + \cdots\right] - \frac{8}{3}\left[1 - \frac{z}{3} + \frac{z^2}{3^2} + \cdots\right]$$

ii) $2 < |z| < 3 \Rightarrow$ $2 < |z| \Rightarrow \frac{2}{|z|} < 1$ & $|z| < 3 \Rightarrow \left|\frac{z}{3}\right| < 1$

$$\therefore f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

$$= 1 + \frac{3}{z\left(1+\frac{2}{z}\right)} - \frac{8}{3\left(1+\frac{z}{3}\right)}$$

$$= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{3}\left(1+\frac{z}{3}\right)^{-1}$$

$$= 1 + \frac{3}{z}\left[1 - \left(\frac{2}{z}\right) + \left(\frac{2}{z}\right)^2 - \left(\frac{2}{z}\right)^3 + \cdots\right] - \frac{8}{3}\left[1 - \frac{z}{3} + \left(\frac{z}{3}\right)^2 - \left(\frac{z}{3}\right)^3 + \cdots\right]$$

iii) $|z| > 3 \Rightarrow 3 < |z|$ $\frac{3}{|z|} < 1 \Rightarrow \left|\frac{3}{z}\right| < 1$

$\Rightarrow \left|\frac{2}{z}\right| < 1$

$$\therefore f(z) = 1 + \frac{3}{z+2} - \frac{8}{z+3}$$

$$= 1 + \frac{3}{z\left(1+\frac{2}{z}\right)} - \frac{8}{z\left(1+\frac{3}{z}\right)}$$

$$= 1 + \frac{3}{z}\left(1+\frac{2}{z}\right)^{-1} - \frac{8}{z}\left(1+\frac{3}{z}\right)^{-1}$$

$$= 1 + \frac{3}{z}\left[1 - \frac{2}{z} + \left(\frac{2}{z}\right)^2 - \left(\frac{2}{z}\right)^3 + \cdots\right] - \frac{8}{z}\left[1 - \frac{3}{z} + \left(\frac{3}{z}\right)^2 - \left(\frac{3}{z}\right)^3 + \cdots\right]$$

<u>Home work</u>

1. Find all Taylor and Laurent's series of $f(z) = \frac{-2z+3}{z^2-3z+2}$ with centre '0'

2. Expand $f(z) = \frac{1}{z^2-4z+3}$, for $1 < |z| < 3$ in Laurent's series.

3. Expand $f(z) = \frac{(z-2)(z+2)}{(z+1)(z+4)}$ in the region

i) $|z| < 1$ ii) $1 < |z| < 4$ iii) $|z| > 4$

Zeros and Singularities:

Zeros: A zero of an analytic function $f(z)$ in a domain D is a point $z = z_0$ such that $f(z_0) = 0$

If $f(z_0) = 0$ & $f'(z_0) \neq 0$, then $z = z_0$ is a simple zero (zero of order 1) of $f(z)$. If $f(z_0) = 0$, $f'(z_0) = 0$ & $f''(z_0) \neq 0$, then $z = z_0$ is a zero of order 2. If $f(z_0) = f'(z_0) = f''(z_0) = \cdots = f^{(n-1)}(z_0) = 0$ and $f^{(n)}(z_0) \neq 0$, then we say ~~that~~ the zero $z = z_0$ is of order n.

eg: i) $f(z) = 1+z^2$ has simple zeros at $z = \pm i$

ii) $f(z) = (1-z^4)^2$ has second order zeros at $z = \pm 1$ and $z = \pm i$

Singularity or Singular point: A function $f(z)$ is singular or has a singularity at a point $z = z_0$ if $f(z)$ is not analytic (perhaps not even defined) at $z = z_0$, but every neighbourhood of $z = z_0$ contain points at which $f(z)$ is analytic.

Isolated Singular point: A point z_0 is an isolated singular point of $f(z)$ if there exist a neighbourhood of z_0 which doesn't contain any other singular point of $f(z)$.

Classification of Isolated Singular points

When $z = z_0$ is an isolated singular point of $f(z)$, we can expand $f(z)$ as a Laurent's series about z_0

ie, $f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \underbrace{\sum_{n=1}^{\infty} b_n (z-z_0)^{-n}}_{\text{(principal part)}}$

valid in a region of the form $0 < |z-z_0| < R$.

Depends on the number of terms in the principal part of $f(z)$ there are three kinds of singularities

i) Removable singularity: In the Laurent's series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$$

if the principal part of $f(z)$ contains no term ie, $b_n=0$ for all n, then the singularity $z=z_0$ is called a removable singularity.

ii) Pole: In the Laurent's series

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$$

if the principal part contains only a finite number of terms, then $z=z_0$ is a pole of the function $f(z)$.

ie, if the principal part of $f(z)$ is

$\frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \cdots + \frac{b_m}{(z-z_0)^m}$ and $b_n = 0$ for all $n>m$.

then $z=z_0$ is called a pole of order 'm'

If $m=1$, then $z=z_0$ is called a simple pole (pole of order 1)

iii) Essential Singularity: If there are infinite number of terms in the principal part ie, if the principal part of $f(z)$ is

$\frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \frac{b_3}{(z-z_0)^3} + \cdots \infty$ then $z=z_0$

is called an essential singularity of $f(z)$.

Residues:

If $f(z)$ has an isolated singularity at a point $z=z_0$, then $f(z)$ has a Laurent's series valid in a region $0<|z-z_0|<R$. The co-efficient of $\frac{1}{z-z_0}$ in the Laurent's series expansion of $f(z)$ about z_0

is called the residue of $f(z)$ at $z = z_0$ and is denoted by $\underset{z=z_0}{\text{Res}}\, f(z)$ or $\text{Res}\{f(z), z=z_0\}$

ie, $\boxed{\underset{z=z_0}{\text{Res}}\, f(z) = \text{co-efficient of } \frac{1}{z-z_0} = b_1}$ from Laurent's series of $f(z)$

Methods for Computing Residues:

i) If $f(z)$ has a simple pole (ie, pole of order 1) at $z=z_0$ then $\text{Res}\{f(z), z=z_0\} = \lim_{z\to z_0} (z-z_0)f(z)$

ii) If $f(z)$ has a pole of order 'm' at $z=z_0$ then

$$\underset{z=z_0}{\text{Res}}\, f(z) = \frac{1}{(m-1)!} \lim_{z\to z_0}\left\{\frac{d^{m-1}}{dz^{m-1}}\left[(z-z_0)^m f(z)\right]\right\}$$

Note: Let $f(z) = \frac{p(z)}{q(z)}$ with $p(z_0) \neq 0$ and $q(z)$ has a simple zero at $z = z_0$, so that $f(z)$ has a simple pole at $z=z_0$. Then

$$\underset{z=z_0}{\text{Res}}\, f(z) = \frac{p(z_0)}{q'(z_0)}$$

Residue Theorem (Cauchy's Residue Theorem)

Let $f(z)$ be analytic inside a simple closed path C and on C, except for finitely many singular points $z_1, z_2, \ldots, z_k$ inside C. Then the integral of $f(z)$ taken counterclockwise around C equals $2\pi i$ times the sum of the residues of $f(z)$ at $z_1, z_2, \ldots, z_k$.

ie, $$\boxed{\oint_C f(z)\,dz = 2\pi i \sum_{j=1}^{k} \underset{z=z_j}{\text{Res}}\, f(z)}$$

[OR $\oint_C f(z)\,dz = 2\pi i \times$ sum of the residues of $f(z)$ at singular points inside C]

Problems

1. Determine and classify the singular points of the following fn

i) $\frac{z-\sin z}{z^3}$ ii) $\frac{e^{-z^2}}{z^2}$ iii) $e^{1/z}$ iv) $z\sin\left(\frac{1}{z}\right)$

Ans: (i) $f(z) = \frac{z-\sin z}{z^3}$

Here $z=0$ is a singular point and isolated.

We obtain the Laurent's series of $f(z)$

$$f(z) = \frac{z-\sin z}{z^3} = \frac{z-\left(z-\frac{z^3}{3!}+\frac{z^5}{5!}-\cdots\right)}{z^3}$$

$\left[\sin z = z-\frac{z^3}{3!}+\frac{z^5}{5!}+\cdots\right]$

$$= \frac{1}{z^3}\left[\frac{z^3}{3!}-\frac{z^5}{5!}+\frac{z^7}{7!}-\cdots\right]$$

$$= \frac{1}{3!}-\frac{z^2}{5!}+\frac{z^4}{7!}-\cdots$$

There are no terms in the principal part → [ie, terms with negative powers of z]

So $z=0$ is a removable singular point.

(ii) $f(z) = \frac{e^{-z^2}}{z^2}$

Here $z=0$ is a singular point and isolated.

$$f(z) = \frac{e^{-z^2}}{z^2} = \frac{1-\frac{z^2}{1!}+\frac{z^4}{2!}-\frac{z^6}{3!}+\frac{z^8}{4!}-\cdots}{z^2}$$

$\left[e^{z} = 1+\frac{z}{1!}+\frac{z^2}{2!}+\cdots \quad e^{-z} = 1-\frac{z}{1!}+\frac{z^2}{2!}-\cdots\right]$

$$= \underbrace{\frac{1}{z^2}}_{\text{principal part}}\underbrace{-1+\frac{z^2}{2!}-\frac{z^4}{3!}+\frac{z^6}{4!}-\cdots}_{\text{Regular part}}$$

There is a single term $\frac{1}{z^2}$ in the principal part of the Laurent's expansion of $f(z)$ at $z=0$

$\therefore$ $z=0$ is a pole of order 2

(iii) $f(z) = e^{1/z}$ $= 1+\frac{1/z}{1!}+\frac{(1/z)^2}{2!}+\frac{(1/z)^3}{3!}+\cdots$

$$= 1+\frac{1}{1!\,z}+\frac{1}{2!\,z^2}+\frac{1}{3!\,z^3}+\cdots$$

There are infinite number of terms in the principal part and so $z=0$ is an essential singularity of $f(z)$

iv) $f(z) = z\sin\left(\frac{1}{z}\right)$

$$= z\left[\frac{1}{z} - \frac{(1/z)^3}{3!} + \frac{(1/z)^5}{5!} - \frac{(1/z)^7}{7!} + \cdots\right]$$

$\left[\because \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right]$

$$= 1 - \frac{1}{3!}\cdot\frac{1}{z^2} + \frac{1}{5!}\cdot\frac{1}{z^4} - \frac{1}{7!}\cdot\frac{1}{z^6} + \cdots$$

There are infinite number of terms in the principal part

$\therefore$ $z=0$ is an essential singularity of $f(z)$

2. Determine and classify the singular points of the following fns

i) $\frac{\sin z}{(z-\pi)^2}$ ii) $\tan z$ iii) $\frac{1}{(z-3)^2(z+5)}$

Ans:

i) $f(z) = \frac{\sin z}{(z-\pi)^2} = \frac{-\sin(z-\pi)}{(z-\pi)^2}$ $[\sin(\pi-\theta) = \sin\theta$

$z=\pi$ is a singular point and isolated. Then

$$\frac{-\sin(z-\pi)}{(z-\pi)^2} = \frac{-\left[(z-\pi) - \frac{(z-\pi)^3}{3!} + \frac{(z-\pi)^5}{5!} - \frac{(z-\pi)^7}{7!} + \cdots\right]}{(z-\pi)^2}$$

$$= -\left[\underbrace{\frac{1}{z-\pi}}_{\text{principal part}} - \frac{(z-\pi)}{3!} + \frac{(z-\pi)^3}{5!} - \frac{(z-\pi)^5}{7!} + \cdots\right]$$

$\Rightarrow$ $z=\pi$ is pole of order 1 (ie, simple pole)

[$\because$ there is a single term in the principal part of $f(z)$ & its power is 1]

ii) $f(z) = \tan z = \frac{\sin z}{\cos z}$

Singular points are given by $\cos z = 0$

$$\Rightarrow z = (2n+1)\pi/2 \; ; \; n = 0, \pm 1, \pm 2, \cdots$$

All singularities are isolated and are simple poles.

(iii) $f(z) = \frac{1}{(z-3)^2(z+5)}$

Singularities are given by $(z-3)^2(z+5) = 0$

$\Rightarrow z = 3$ and $z = -5$

$z = 3$ is a pole of order 3

$z = -5$ is a pole of order 1

3. Find the singular points and residue of the following fns:

i) $\frac{1-e^{2z}}{z^4}$ ii) $\frac{\sinh z}{z^4}$

Ans:

(i) $f(z) = \frac{1-e^{2z}}{z^4}$ [Here $z=0$ is a singular point]

We obtain the Laurent's expansion of $f(z)$ about $z=0$

$$= \frac{1-\left[1+\frac{(2z)}{1!}+\frac{(2z)^2}{2!}+\frac{(2z)^3}{3!}+\frac{(2z)^4}{4!}+\cdots\right]}{z^4} \quad \left[\because e^z = 1+\frac{z}{1!}+\frac{z^2}{2!}+\cdots\right]$$

$$= \underbrace{\frac{-2}{z^3}-\frac{2}{z^2}-\frac{4}{3z}}_{\text{principal part}}-\frac{2}{3}-\frac{4}{15}z-\cdots$$

Since the highest power of $\frac{1}{z-0}$ is 3,

$z=0$ is a pole of order 3

Residue:
$=$ co-efficient of $\frac{1}{z-z_0}$
$=$ co-efficient of $\frac{1}{z-0}$
$=$ co-efficient of $\frac{1}{z}$

$\therefore \operatorname{Res}_{z=0} f(z) =$ co-efficient of $\frac{1}{z-0} = \frac{1}{z} = -\frac{4}{3}$

(ii) $f(z) = \frac{\sinh z}{z^4}$ (Here $z=0$ is a singular point) $= \frac{z+\frac{z^3}{3!}+\frac{z^5}{5!}+\cdots}{z^4}$

$$= \underbrace{\frac{1}{z^3}+\frac{1}{3!\,z}}_{\text{principal part}}+\frac{z}{5!}+\cdots$$

Laurent's expansion of $f(z)$ about $z=0$

$\therefore z=0$ is a pole of order 3

$\operatorname{Res}_{z=0} f(z) =$ co-efficient of $\frac{1}{z-0}$ (in the Laurent's expansion of $f(z)$) $= \frac{1}{z} = \frac{1}{3!} = \frac{1}{6}$

4. Find all singularities and residues of the following fns

i) $\dfrac{ze^z}{z^2+4}$ ii) $\tan z$ iii) $\dfrac{z^2-z}{(z+1)^2(z^2+4)}$

Ans: (i) $f(z) = \dfrac{ze^z}{z^2+4}$

Singular points are given by $z^2+4=0 \Rightarrow z^2=-4$

$\Rightarrow z = \pm 2i$

All singularities are simple poles.

$$\left[\because f(z) = \frac{ze^z}{z^2+4} = \frac{ze^z}{(z+2i)(z-2i)}\right]$$

$$\operatorname*{Res}_{z=2i} f(z) = \lim_{z\to 2i} (z-2i)f(z)$$
$$= \lim_{z\to 2i} (z-2i)\frac{ze^z}{(z+2i)(z-2i)}$$
$$= \lim_{z\to 2i} \frac{ze^z}{z+2i} = \frac{2ie^{2i}}{2i+2i} = \frac{2ie^{2i}}{4i} = \underline{\underline{\frac{e^{2i}}{2}}}$$

$$\operatorname*{Res}_{z=-2i} f(z) = \lim_{z\to -2i} (z+2i)f(z)$$
$$= \lim_{z\to -2i} (z+2i)\frac{ze^z}{(z+2i)(z-2i)}$$
$$= \lim_{z\to -2i} \frac{ze^z}{z-2i} = \frac{-2ie^{-2i}}{-2i-2i} = \frac{-2ie^{-2i}}{-4i} = \underline{\underline{\frac{e^{-2i}}{2}}}$$

(ii) $f(z) = \tan z$

The singular points of $f(z) = \tan z = \dfrac{\sin z}{\cos z}$ are given by

$\cos z = 0 \Rightarrow z = (2n+1)\pi/2$; $n = 0, \pm 1, \pm 2, \cdots$

All singularities are simple poles.

$$\operatorname*{Res}_{z=(2n+1)\pi/2} f(z) = \lim_{z\to(2n+1)\pi/2} \left[z-(2n+1)\pi/2\right]\frac{\sin z}{\cos z}$$
$$= \lim_{z\to(2n+1)\pi/2} \frac{\left[z-(2n+1)\pi/2\right]\sin z}{\cos z} \quad \left(\tfrac{0}{0} \text{ form}\right)$$

$$= \lim_{z \to (2n+1)\pi/2} \frac{\cos z\,[z-(2n+1)\pi/2] + \sin z \cdot 1}{-\sin z}$$ [L' Hospital's Rule]

$$= \lim_{z \to (2n+1)\pi/2} \frac{0+\sin(2n+1)\pi/2}{-\sin(2n+1)\pi/2} = \underline{\underline{-1}}$$

OR The singular points of $f(z) = \tan z = \frac{\sin z}{\cos z}$ are

$$z = (2n+1)\pi/2 \quad ; n = 0, \pm 1, \pm 2, \dots$$

All singularities are simple poles

$$\therefore \operatorname*{Res}_{z=(2n+1)\pi/2} f(z) = \frac{p(z_0)}{q'(z_0)} = \left.\frac{\sin z}{-\sin z}\right]_{z_0=(2n+1)\pi/2}$$

$$= \underline{\underline{-1}}$$

$f(z) = \frac{\sin z}{\cos z} = \frac{p(z)}{q(z)}$ where $p(z_0) \neq 0$ & $q(z)$ has a simple zero at z_0

(iii) $f(z) = \dfrac{z^2 - z}{(z+1)^2(z^2+4)}$

The singular points are given by $(z+1)^2(z^2+4) = 0$

$$\Rightarrow (z+1)^2 = 0 \;;\; z^2+4 = 0$$

$$\Rightarrow z = -1, \quad z = \pm 2i$$

$z = -1$ is a pole of order 2 and $z = \pm 2i$ are simple poles

$$\operatorname*{Res}_{z=2i} f(z) = \lim_{z\to 2i}(z-2i)\, f(z) = \lim_{z \to 2i} (z-2i) \frac{z^2-z}{(z+1)^2(z^2+4)}$$

$$= \lim_{z\to 2i} (z-2i) \frac{z^2-z}{(z+1)^2(z-2i)(z+2i)}$$

$$= \lim_{z\to 2i} \frac{z^2-z}{(z+1)^2(z+2i)}$$

$$= \frac{(2i)^2 - 2i}{(2i+1)^2(4i)} = \frac{-4-2i}{(4i-3)\,4i}$$

$$= \frac{-2(2+i)}{-(3-4i)\,4i} = \frac{2+i}{2i(3-4i)}$$

$$= \frac{2+i}{2(4+3i)}$$

$$\operatorname*{Res}_{z=-2i} f(z) = \lim_{z\to -2i} (z+2i) f(z) = \lim_{z\to -2i} (z+2i) \frac{z^2-z}{(z+1)^2(z+2i)(z-2i)}$$

$$= \lim_{z\to -2i} \frac{z^2-z}{(z+1)^2(z-2i)} = \frac{(-2i)^2-(-2i)}{(-2i+1)^2(-4i)}$$

$$= \frac{-4+2i}{-4i(-3-4i)} = \frac{2(i-2)}{4i(3+4i)} = \underline{\frac{i-2}{2(3i-4)}}$$

$$\operatorname*{Res}_{z=-1} f(z) = \frac{1}{(2-1)!} \lim_{z\to -1} \left\{ \frac{d}{dz} \left[(z+1)^2 f(z) \right] \right\}$$

[Here $z=-1$ is a pole of order $m=1$]

$$= \frac{1}{1!} \lim_{z\to -1} \left\{ \frac{d}{dz} \left[(z+1)^2 \frac{z^2-z}{(z+1)^2(z^2+4)} \right] \right\}$$

$$= \lim_{z\to -1} \left\{ \frac{d}{dz} \left[\frac{z^2-z}{z^2+4} \right] \right\}$$

$$= \lim_{z\to -1} \left\{ \frac{(z^2+4)(2z-1)-(z^2-z)2z}{(z^2+4)^2} \right\}$$

$$= \frac{(5)(-3)+4}{5^2} = \underline{\underline{\frac{-11}{25}}}$$

5 Using Residue theorem evaluate the following (counterclockwise)

(i) $\oint_C z^4 e^{1/z^2} dz$ where C is $|z|=1$

Ans: Let $f(z) = z^4 e^{1/z^2}$

$$= z^4\left[1 + \frac{1/z^2}{1!} + \frac{(1/z^2)^2}{2!} + \frac{(1/z^2)^3}{3!} + \frac{(1/z^2)^4}{4!} + \cdots\right]$$

$$= z^4\left[1 + \frac{1}{1!z^2} + \frac{1}{2!z^4} + \frac{1}{3!z^6} + \frac{1}{4!z^8} + \cdots\right]$$

$$= z^4 + \frac{z^2}{1!} + \frac{1}{2!} + \frac{1}{3!z^2} + \frac{1}{4!z^4} + \cdots \quad (1)$$

$\therefore z=0$ is a singular point (essential singular point) of $f(z)$ lies inside $|z|=1$

$\operatorname*{Res}_{z=0} f(z)$ = Co-efficient of $\frac{1}{z-0} = \frac{1}{z}$ in the Laurent's expansion (1) of $f(z)$

$= 0$

$\therefore$ by Cauchy's Residue theorem, $\oint_C z^4 e^{1/z^2} dz = 2\pi i \operatorname*{Res}_{z=0} f(z)$

$= 2\pi i(0) = \underline{\underline{0}}$

(ii) $\oint_C \frac{\sin z}{z^4}\, dz$ where C is $|z|=3$

Ans: Let $f(z) = \frac{\sin z}{z^4}$

$z=0$ is a singular point of $f(z)$ which lies inside C

$$f(z) = \frac{\sin z}{z^4} = \frac{1}{z^4}\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots\right]$$

$$= \underbrace{\frac{1}{z^3} - \frac{1}{3!\,z}}_{\text{principal part}} + \frac{z}{5!} - \frac{z^3}{7!} + \cdots \quad \text{---} (1)$$

$\Rightarrow z=0$ is a pole of order 3

$\therefore \operatorname{Res}_{z=0} f(z) = $ co-efficient of $\frac{1}{z-0}$ in (1)

$= -\frac{1}{3!}$

$= -1/6$

$\therefore \oint_C \frac{\sin z}{z^4}\, dz = 2\pi i \times$ sum of residues at singular points within C

$= 2\pi i \left[\operatorname{Res}_{z=0} f(z)\right] = 2\pi i\left(-\frac{1}{6}\right) = \frac{-\pi i}{3}$

(iii) $\oint_C \frac{dz}{z^2(z-1)}$ where C is $|z|=2$

Ans: Let $f(z) = \frac{1}{z^2(z-1)}$

The singular points of $f(z)$ are given by

$z^2(z-1) = 0 \Rightarrow z = 0, 1$

Both the singular points $z=0,1$ lie inside $C : |z|=2$

$z=1$ is a simple pole & $z=0$ is a pole of order 2.

$$\operatorname{Res}_{z=1} f(z) = \lim_{z\to 1} (z-1) f(z) = \lim_{z\to 1} (z-1)\frac{1}{z^2(z-1)}$$

$$= \lim_{z\to 1} \frac{1}{z^2} = 1$$

$$\operatorname*{Res}_{z=0} f(z) = \frac{1}{(2-1)!} \lim_{z\to 0} \left\{ \frac{d}{dz} \left[(z-0)^2 \frac{1}{z^2(z-1)} \right] \right\}$$ $\left[\begin{array}{l} z=0 \text{ is a} \\ \text{pole of order} \\ m=2 \end{array} \right.$

$$= \lim_{z\to 0} \left\{ \frac{d}{dz} \left[\frac{1}{z-1} \right] \right\}$$

$$= \lim_{z\to 0} \left[\frac{-1}{(z-1)^2} \right] = \underline{-1}$$

By Residue theorem, $\oint_C \frac{1}{z^2(z-1)} dz = 2\pi i \times$ sum of residues at singular points within C

$$= 2\pi i \left[\operatorname*{Res}_{z=1} f(z) + \operatorname*{Res}_{z=0} f(z) \right]$$

$$= 2\pi i \left[1 - 1 \right]$$

$$= \underline{\underline{0}}$$

(iv) $\oint_C \frac{\sin^2 z}{(z-\pi/6)^3} dz$ where C is $|z|=1$

<u>Ans:</u> Let $f(z) = \frac{\sin^2 z}{(z-\pi/6)^3}$

Singular point of $f(z)$ are given by $(z-\pi/6)^3 = 0$

$\Rightarrow z = \pi/6$ is a pole of order 3

The point $z=\pi/6$ lies inside the curve $C: |z|=1$

$\therefore$ by Residue theorem,

$$\oint_C \frac{\sin^2 z}{(z-\pi/6)^3} dz = 2\pi i \left[\operatorname*{Res}_{z=\pi/6} f(z) \right] \quad \text{——— (1)}$$

$$\operatorname*{Res}_{z=\pi/6} f(z) = \frac{1}{(3-1)!} \lim_{z\to\pi/6} \left\{ \frac{d^2}{dz^2} \left[(z-\pi/6)^3 \frac{\sin^2 z}{(z-\pi/6)^3} \right] \right\}$$ $\left[\begin{array}{l} \text{Here} \\ m=3 \end{array} \right]$

$$= \frac{1}{2} \lim_{z\to\pi/6} \left\{ \frac{d^2}{dz^2} \left[\sin^2 z \right] \right\}$$

$$= \frac{1}{2} \lim_{z\to\pi/6} \left\{ \frac{d}{dz} \left[2\sin z \cos z \right] \right\}$$

Result: $2\sin\theta\cos\theta = \sin 2\theta$

$$= \frac{1}{2} \lim_{z\to\pi/6} \left\{ \frac{d}{dz} \left[\sin 2z \right] \right\}$$

$$= \frac{1}{2} \lim_{z\to\pi/6} \left[2\cos 2z \right] = \frac{1}{2}(1) = \underline{1/2}$$

$\therefore \oint_C \frac{\sin^2 z}{(z-\pi/6)^3} dz = 2\pi i \left[\frac{1}{2}\right] = \underline{\underline{\pi i}}$ from (1)

(v) $\oint_C \frac{z^2}{(z-1)^2(z+2)} dz$ where C is $|z-2| = 2$

<u>Ans:</u> $f(z) = \frac{z^2}{(z-1)^2(z+2)}$

The singular points of $f(z)$ are given by

$(z-1)^2(z+2) = 0 \implies (z-1)^2 = 0 \quad z+2 = 0$

$\implies z = 1 \ ; \ z = -2$

The singular point $z = -2$ lies <u>outside</u> $C : |z-2| = 2$

$\left[\because z = -2 \implies |z-2| = |-2-2| = |-4| = 4 > 2\right]$

The singular point $z = 1$ lies <u>inside C</u> : $|z-2| = 2$

$\left[\because z = 1 \implies |z-2| = |1-2| = |-1| = 1 < 2\right]$

$z = 1$ is a pole of order 2 $\left[\because f(z) = \frac{z^2}{(z-1)^2(z+2)}\right.$

$\therefore$ by Cauchy's Residue theorem,

$$\oint_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\operatorname{Res}_{z=1} f(z)\right] \quad \text{——— (1)}$$

$$\operatorname{Res}_{z=1} f(z) = \frac{1}{(2-1)!} \lim_{z\to 1} \left\{\frac{d}{dz}\left[(z-1)^2 f(z)\right]\right\}$$

$$= \frac{1}{1!} \lim_{z\to 1} \left\{\frac{d}{dz}\left[(z-1)^2 \frac{z^2}{(z-1)^2(z+2)}\right]\right\}$$

$$= \lim_{z\to 1}\left\{\frac{d}{dz}\left[\frac{z^2}{z+2}\right]\right\} = \lim_{z\to 1} \frac{(z+2)(2z) - z^2(1)}{(z+2)^2}$$

$$= \frac{(3)(2) - 1}{3^2} = \frac{5}{9}$$

$$\therefore \oint_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i \left[\frac{5}{9}\right] = \underline{\underline{\frac{10\pi i}{9}}}$$

(vi) $\oint_C \frac{z-23}{z^2-4z-5}$ where $C : |z-2-i| = 3.2$.

Ans: $f(z) = \frac{z-23}{z^2-4z-5} = \frac{z-23}{(z-5)(z+1)}$

The singular points of $f(z)$ are given by

$(z-5)(z-1) = 0 \implies z = 5, -1$

Both the points $z=5$ & $z=-1$ lie inside C

$\left[\because z=5 \implies |5-2-i| = |3-i| = \sqrt{3^2+(-1)^2} = \sqrt{10} = 3.16 < 3.2 \right.$

$z=-1 \implies |-1-2-i| = |-3-i| = \sqrt{(-3)^2+(-1)^2} = \sqrt{10} = 3.16 < 3.2$

$z=5$ & $z=-1$ are simple poles $\left[\because f(z) = \frac{z-23}{(z-5)(z+1)} \right]$

$\operatorname{Res}_{z=5} f(z) = \lim_{z\to 5} (z-5) f(z) = \lim_{z\to 5} (z-5) \frac{(z-23)}{(z-5)(z+1)}$

$= \lim_{z\to 5} \frac{z-23}{z+1} = \frac{-18}{6} = -3$

$\operatorname{Res}_{z=-1} f(z) = \lim_{z\to -1} (z+1) \frac{(z-23)}{(z-5)(z+1)} = \lim_{z\to -1} \frac{z-23}{z-5}$

$= \frac{-24}{-6} = 4$

$\therefore$ by Cauchy's Residue theorem,

$\oint_C \frac{z-23}{z^2-4z-5} dz = 2\pi i \left[\operatorname{Res}_{z=5} f(z) + \operatorname{Res}_{z=-1} f(z) \right]$

$= 2\pi i \left[-3 + 4 \right] = \underline{2\pi i}$

Home work : Using Residue theorem evaluate

1. $\oint_C \frac{\tan z}{z^2-1} dz$ counterclockwise around $C : |z| = 3/2$

2. $\oint_C \frac{e^z}{\sin z}$ where C is $|z|=1$ counterclockwise

3. $\oint_C \frac{30z^2 - 23z + 5}{(2z-1)^2(3z-1)} dz$ where C is $|z|=1$ counterclockwise

4. Integrate $f(z) = \frac{1}{z^3 - z^4}$ clockwise around the circle $C: |z| = \frac{1}{2}$

5. Evaluate $\oint_C \left(\frac{z \cdot e^{\pi z}}{z^4 - 16} + z e^{\pi z} \right) dz$ where C is the ellipse $9x^2 + y^2 = 9$, counterclockwise.

6. Find the poles and residues of the function

$$f(z) = \frac{z^2 - 2z}{(z+1)^2 (z^2+4)}$$

Contour Integration.

I) Integrals of Rational functions of Sin θ and Cos θ.

$$\int_0^{2\pi} f(\sin\theta, \cos\theta)\, d\theta .$$

To evaluate put $z = e^{i\theta} = \cos\theta + i\sin\theta$

$$\frac{1}{z} = e^{-i\theta} = \cos\theta - i\sin\theta$$

$$z + \frac{1}{z} = 2\cos\theta \Rightarrow \cos\theta = \frac{1}{2}\left[z + \frac{1}{z}\right]$$

$$z - \frac{1}{z} = 2i\sin\theta \Rightarrow \sin\theta = \frac{1}{2i}\left[z - \frac{1}{z}\right]$$

$z = e^{i\theta}$ $\qquad$ $\theta = 0 \Rightarrow z = 1$

$dz = e^{i\theta} \cdot i\, d\theta.$ $\qquad$ $\theta = 2\pi \Rightarrow z = 1$

$= z\, i\, d\theta$ $\qquad$ $\therefore$ C is a circle with r=1

$\therefore d\theta = \frac{dz}{iz}$ $\qquad$ $|z| = 1.$

$$\therefore \int_0^{2\pi} f(\sin\theta, \cos\theta)\, d\theta$$

$$= \int_{|z|=1} f\left(\frac{1}{2i}\left(z - \frac{1}{z}\right), \frac{1}{2}\left(z + \frac{1}{z}\right)\right) \frac{dz}{iz}$$

$$= \int_C f(z)\, dz \text{ ., when C is } |z| = 1.$$

By using Residue Theorem

$$= 2\pi i \left[\text{Sum of the residues at poles inside } |z| = 1\right]$$

① Evaluate $\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta}$.

$z = e^{i\theta} = \cos\theta + i\sin\theta$

$\frac{1}{z} = e^{-i\theta} = \cos\theta - i\sin\theta$

$\cos\theta = \frac{1}{2}\left[z + \frac{1}{z}\right]$

$z = e^{i\theta}$

$dz = i e^{i\theta} d\theta$

$= i z \, d\theta$

$d\theta = \frac{dz}{iz}$

$c : |z| = 1$ [Integration over a unit circle]

$$\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta} = \int_C \frac{dz/iz}{5 - \frac{3}{2}\left[z + \frac{1}{z}\right]} \quad , \text{ when } c: |z|=1$$

$$= \frac{1}{i}\int_C \frac{dz}{z\left[\frac{10 - 3\left[\frac{z^2+1}{z}\right]}{2}\right]}$$

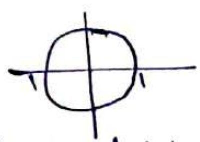
$$= \frac{2}{i}\int_C \frac{dz}{z\left[\frac{10z - 3z^2 - 3}{z}\right]}$$

$$= \frac{2}{i}\int_C \frac{dz}{-3z^2 + 10z - 3}$$

$$= \frac{-2}{i}\int_C \frac{dz}{3z^2 - 10z + 3}$$

$$= \frac{-2}{i}\int_C \frac{dz}{3\left[z-3\right]\left[z-\frac{1}{3}\right]}$$

$$= \frac{-2}{3i}\int_C \frac{dz}{(z-3)\left(z-\frac{1}{3}\right)}$$



Singularities are $z = 3$ and $z = \frac{1}{3}$,

$z = \frac{1}{3}$ lies inside $|z|=1$

$z = 3$ lies outside $|z|=1$

$$= \frac{-2}{3i}\left[2\pi i\left[\text{Res } f(z)\right]_{\text{at } z=\frac{1}{3}}\right] \longrightarrow (1) \quad \text{[using Residue Theorem]}$$

$$\left[\text{Res } f(z)\right]_{z=\frac{1}{3}} = \lim_{z\to\frac{1}{3}}\left[\left(z-\tfrac{1}{3}\right)\frac{1}{(z-3)\left(z-\frac{1}{3}\right)}\right]$$

$$= \lim_{z\to\frac{1}{3}}\left[\frac{1}{z-3}\right] = \frac{1}{\frac{1}{3}-3}$$

$$= \frac{1}{\frac{1-9}{3}} = -\frac{3}{8}$$

$$(1) \Rightarrow \int_0^{2\pi}\frac{d\theta}{5-3\cos\theta} = -\frac{2}{3i}\left[2\pi i \times -\frac{3}{8}\right]$$

$$= \frac{\pi}{2}$$

(2) Using contour integration, Evaluate $\int_0^{2\pi}\frac{1}{5-3\sin\theta}\,d\theta$.

Consider the unit circle $|z|=1$

$$z = e^{i\theta}, \quad 0 \le \theta \le 2\pi$$

$$z = e^{i\theta} = \cos\theta + i\sin\theta$$

$$\frac{1}{z} = e^{-i\theta} = \cos\theta - i\sin\theta$$

$$z - \frac{1}{z} = 2i\sin\theta$$

$$\therefore \sin\theta = \frac{1}{2i}\left[z-\frac{1}{z}\right] = \frac{z^2-1}{2iz}.$$

$$z = e^{i\theta} \Rightarrow dz = ie^{i\theta}\,d\theta.$$

$$= iz\,d\theta$$

$$\therefore d\theta = \frac{dz}{iz}.$$

$$\therefore \int_0^{2\pi} \frac{d\theta}{5-3\sin\theta} = \int_{|z|=1} \frac{1}{5-3\left[\frac{z^2-1}{2iz}\right]} \times \frac{dz}{iz}.$$

$$= \int_{|z|=1} \frac{2iz}{10iz-3z^2+3} \times \frac{dz}{iz}.$$

$$= \int_C \frac{2\,dz}{-3z^2+10iz+3} \quad \text{, when } C \Rightarrow |z|=1$$

Singular points are given by $-3z^2+10iz+3=0$

$$z = 3i,\ \frac{i}{3}.$$

$z = 3i \rightarrow (0,3)$ lies outside $|z|=1$.

$z = \frac{i}{3} \Rightarrow (0, \frac{1}{3})$ lies inside $|z|=1$. (Simple pole)

$$= 2\int_C \frac{dz}{-3z^2+10iz+3}$$

$$= 2\int_C \frac{dz}{-3\left[z-3i\right]\left[z-\frac{i}{3}\right]}$$ [Apply Residue Theorem].

$$= \frac{2}{-3}\left[2\pi i \left[\text{Sum of the residues at poles inside } |z|=1\right]\right.$$

$$= \frac{2}{-3}\left[2\pi i\left[\text{Res } f(z) \text{ at } z=\frac{i}{3}\right]\right.$$

$$= \frac{4\pi i}{-3}\left[\lim_{z\to\frac{i}{3}} \left(z-\frac{i}{3}\right)\frac{1}{(z-3i)\left(z-\frac{i}{3}\right)}\right]$$

$$= \frac{4\pi i}{-3}\left[\frac{1}{\frac{i}{3}-3i}\right] = \frac{4\pi i}{-3}\left[\frac{3}{-8i}\right] = \frac{\pi}{2}$$

(3) Using contour integration, Evaluate

$$\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta.$$

Consider $|z|=1$, $z=e^{i\theta}$, $0 \le \theta \le 2\pi$.

$z = e^{i\theta} = \cos\theta + i\sin\theta$

$\frac{1}{z} = e^{-i\theta} = \cos\theta - i\sin\theta$

$z + \frac{1}{z} = 2\cos\theta$ $\quad \therefore \cos\theta = \frac{z+\frac{1}{z}}{2} = \frac{z^2+1}{2z}$.

$dz = e^{i\theta} \cdot i\,d\theta$

$= z\,i\,d\theta$

$d\theta = \frac{dz}{iz}$

$z^2 = e^{i2\theta} = \cos 2\theta + i\sin 2\theta$

$\frac{1}{z^2} = e^{-i2\theta} = \cos 2\theta - i\sin 2\theta$

$z^2 + \frac{1}{z^2} = 2\cos 2\theta$

$$\cos 2\theta = \frac{1}{2}\left[z^2 + \frac{1}{z^2}\right]$$

$$= \frac{z^4+1}{2z^2}$$

$$\therefore \int_0^{2\pi} \left(\frac{\cos 2\theta}{5+4\cos\theta}\right) d\theta$$

$$= \int_{|z|=1} \frac{\frac{z^4+1}{2z^2}}{5+4\left[\frac{z^2+1}{2z}\right]} \times \frac{dz}{iz}$$

$$= \int_C \frac{z^4+1}{2z^2\left[5+2z+\frac{2}{z}\right]} \times \frac{dz}{iz}$$

$$= \oint_C \frac{(z^4+1)z}{2z^2[5z+2z^2+2]} \times \frac{dz}{iz}. \quad , \text{ when } C \text{ is } |z|=1$$

$$= \frac{1}{2i} \oint_C \frac{z^4+1}{z^2(2z^2+5z+2)} \, dz.$$

$$= \frac{1}{2i} \oint_C \frac{z^4+1}{z^2 \times 2\left[z+\frac{1}{2}\right][z+2]} \, dz$$

$$= \frac{1}{4i} \oint_C \frac{z^4+1}{z^2\left[z+\frac{1}{2}\right][z+2]} \, dz. \quad — (1) \left\{ \oint_C f(z)\,dz \text{ form} \right]$$

Apply Residue Theorem for

$$f(z) = \frac{z^4+1}{z^2\left[z+\frac{1}{2}\right][z+2]}, \quad C: |z|=1$$

$z=0$ is a pole of order 2 lies inside $|z|=1$

$z=-\frac{1}{2}$ is a simple pole lies inside $|z|=1$.

$z=-2$ is a simple pole lies outside $|z|=1$.

(2)

$$(1) \Rightarrow = \frac{1}{4i}\left[2\pi i\left[\left[\text{Res } f(z)\right]_{z=0} + \left[\text{Res } f(z)\right]_{z=-\frac{1}{2}}\right]\right]$$

$$\left[\text{Res } f(z)\right]_{z=0} = \lim_{z\to 0} \frac{1}{(2-1)!} \frac{d}{dz} z^2\left[\frac{z^4+1}{z^2(z+\frac{1}{2})(z+2)}\right]$$

$$= \lim_{z\to 0} \frac{d}{dz}\left[\frac{z^4+1}{(z+\frac{1}{2})(z+2)}\right]$$

$$= \lim_{z\to 0}\left[\frac{(z+\frac{1}{2})(z+2)(4z^3) - (z^4+1)\left[(z+\frac{1}{2})+z+2\right]}{\left[(z+\frac{1}{2})(z+2)\right]^2}\right]$$

$$= \left[\frac{0 - \left(\frac{1}{2}+2\right)}{\left[\frac{1}{2}\times 2\right]^2}\right] = \frac{-5}{2}$$

$$\left[\text{Res } f(z)\right]_{z=-\frac{1}{2}} = \lim_{z\to -\frac{1}{2}} \cancel{\left(z+\frac{1}{2}\right)} \left[\frac{z^4+1}{z^2\cancel{\left(z+\frac{1}{2}\right)}(z+2)}\right]$$

$$= \lim_{z\to -\frac{1}{2}} \left[\frac{z^4+1}{z^2(z+2)}\right]$$

$$= \frac{\frac{1}{16}+1}{\frac{1}{4}\left(\frac{-1}{2}+2\right)} = \frac{17/16}{\frac{1}{4}\left[\frac{3}{2}\right]} = \frac{17}{16}\times\frac{8}{3}$$

$$= \frac{17}{6}$$

Substituting Residues in (2)

$$\int_0^{2\pi}\left(\frac{\cos 2\alpha}{5+4\cos\alpha}\right)d\alpha$$

$$= \frac{2\pi i}{4i}\left[-\frac{5}{2}+\frac{17}{6}\right]$$

$$= \frac{\pi}{2}\left[\frac{1}{3}\right] = \frac{\pi}{6}$$

(4) Solve by using Residues $\int_0^{2\pi} \frac{\cos 2\alpha}{1-2a\cos\alpha+a^2}\,d\alpha$

when $a^2 < 1$.

~~$|z|=1$~~ $z = e^{i\alpha} = \cos\alpha + i\sin\alpha$

$$\frac{1}{z} = \cos\alpha - i\sin\alpha$$

$$\cos\theta = \frac{z^2+1}{2z}, \quad \cos 2\theta = \frac{z^2+\frac{1}{z^2}}{2}$$

$$= \frac{z^4+1}{2z^2}$$

$$z = e^{i\theta}$$

$$dz = e^{i\theta}.\, i\, d\theta$$

$$= z\, i\, d\theta$$

$$d\theta = \frac{dz}{iz}.$$

$$\int_0^{2\pi}\left(\frac{\cos 2\theta}{1-2a\cos\theta+a^2}\right)d\theta$$

$$= \int_C \frac{\frac{z^4+1}{2z^2}}{1-2a\left(\frac{z^2+1}{2z}\right)+a^2} \times \frac{dz}{iz}, \text{ when } C \text{ is } |z|=1.$$

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$$= \int_C \frac{z^4+1}{2z^2\left[\frac{z-az^2-a+a^2z}{z}\right]} \times \frac{dz}{iz}$$

$$= \int_C \frac{(z^4+1)\cancel{z}}{2z^2(-az^2+z(1+a^2)-a)} \times \frac{dz}{i\cancel{z}}$$

$$= \frac{1}{2i}\int_C \frac{z^4+1}{z^2(z-a)(1-az)}\, dz.$$

$$-az^2+z(1+a^2)-a$$
$$= z-az^2-a+a^2z$$
$$= z-a+az(a-z)$$
$$= z-a\,[1-az)]$$

$$1-az=0$$
$$az=1$$
$$z=\frac{1}{a}.$$

$z=0$ pole of order 2 } lies inside C

$z=a$ pole of order 1 }

$z=\frac{1}{a}>1$ lies outside C.

$$= \frac{1}{2i}\left[2\pi i\left[\left[\text{Res } f(z)\right]_{z=0} + \left[\text{Res } f(z)\right]_{z=a}\right]\right]. \quad — (1)$$

$$\left[\text{Res } f(z)\right]_{z=0} = \lim_{z\to 0} \frac{1}{(2-1)!} \frac{d}{dz} \cancel{z^2} \left[\frac{z^4+1}{\cancel{z^2}(z-a)(1-az)}\right]$$

$$= \lim_{z\to 0} \frac{d}{dz}\left[\frac{z^4+1}{(z-a)(1-az)}\right]$$

$$= \lim_{z\to 0}\left[\frac{(z-a)(1-az)(4z^3) - (z^4+1)\left[(z-a)(-a)+(1-az)\right]}{\left[(z-a)(1-az)\right]^2}\right]$$

$$= \frac{0-(a^2+1)}{a^2} = \frac{-(1+a^2)}{a^2}$$

$$\left[\text{Res } f(z)\right]_{z=a} = \lim_{z\to a} (z-a)\left[\frac{(z^4+1)}{z^2(z-a)(1-az)}\right]$$

$\{z=a$ pole of order $1\}$

$$= \lim_{z\to a}\left[\frac{z^4+1}{z^2(1-az)}\right]$$

$$= \frac{a^4+1}{a^2(1-a^2)} = \frac{1+a^4}{a^2-a^4}$$

(1) $\Rightarrow$

$$\int_0^{2\pi}\left(\frac{\cos 2\theta}{1-2a\cos\theta+a^2}\right)d\theta = \frac{1}{2i}\left[2\pi i\left[-\frac{(1+a^2)}{a^2} + \frac{1+a^4}{a^2-a^4}\right]\right]$$

$$= \pi\left[\frac{-1-a^2}{a^2} + \frac{a^4+1}{a^2(1-a^2)}\right]$$

$$= \frac{\pi}{a^2}\left[-1-a^2+\frac{a^4+1}{1-a^2}\right]$$

$$= \frac{\pi}{a^2}\left[\frac{(1-a^2)(-1-a^2)+a^4+1}{1-a^2}\right]$$

$$= \frac{\pi}{a^2}\left[\frac{-1+a^2-a^2+a^4+a^4+1}{1-a^2}\right]$$

$$= \frac{\pi}{a^2}\left[\frac{2a^4}{1-a^2}\right] = \frac{2\pi a^2}{1-a^2}$$

H.W

(1) Evaluate

(a) $\displaystyle\int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$

(b) $\displaystyle\int_0^{2\pi} \left(\frac{\sin^2\theta}{2+\cos\theta}\right) d\theta$

Note $\quad z = e^{i\theta}$

$$z^2 = e^{i2\theta} = \cos 2\theta + i\sin 2\theta$$

$$\frac{1}{z^2} = e^{-i2\theta} = \cos 2\theta - i\sin 2\theta$$

$$z^2 - \frac{1}{z^2} = 2i\sin 2\theta$$

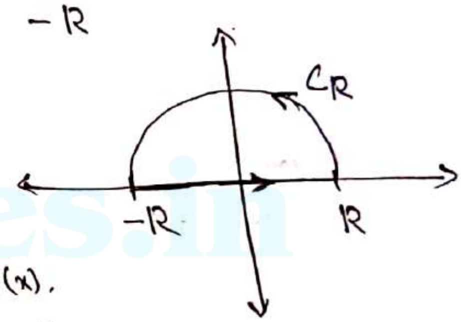
$$\therefore \sin 2\theta = \frac{1}{2i}\left[z^2 - \frac{1}{z^2}\right]$$

Contour Integration around a Semi-Circle.

Evaluation of integrals of the form $\int_{-\infty}^{\infty} f(x)\,dx$ where $f(x)$ has no poles on the real axis.

Consider $\int_C f(z)dz$, where C is the closed contour consisting of the upper half of the circle $|z|=R$ (denoted as C_R) and the real axis from $-R$ to R.

$$\int_C f(z)\,dz = \int_{|z|=R} f(z)dz + \int_{-R}^{R} f(z)\,dz.$$



Along $-R$, to R, part of the real axis, when $y=0$

$\therefore z = x+iy = x.$ $\quad \therefore f(z) = f(x).$

$$\therefore \int_C f(z)\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx.$$

when $R \to \infty$

$$\int_C f(z)dz = \int_{C_R} f(z)dz + \int_{-\infty}^{\infty} f(x)\,dx. \quad \text{——} \quad (1)$$

when $R\to\infty$, we can prove that

$$\int_{C_R} f(z)\,dz \to 0.$$

$$\therefore (1) \Rightarrow \int_C f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx.$$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_C f(z)\,dz$$

$$= 2\pi i \left[\text{Sum of the residues at poles inside } C\right].$$ [using Residue Theorem]

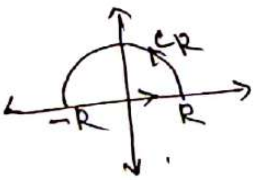
① Evaluate $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx$ using contour integration.

$$f(x) = \frac{x^2}{(x^2+1)(x^2+4)} \qquad f(z) = \frac{z^2}{(z^2+1)(z^2+4)}$$

$z = \pm i, \pm 2i$ are the poles.

Consider $\int_C f(z)\,dz$, where C is the upper half of the circle $|z| = R$ and the real axis from $-R$ to R.

$$\therefore \int_C \frac{z^2}{(z^2+1)(z^2+4)}\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx.$$



$$= \int_{C_R} \frac{z^2}{(z^2+1)(z^2+4)}\,dz + \int_{-R}^{R} \frac{x^2}{(x^2+1)(x^2+4)}\,dx$$

when $R \to \infty$,

$$\int_C \frac{z^2}{(z^2+1)(z^2+4)}\,dz = \int_{C_R} \frac{z^2}{(z^2+1)(z^2+4)}\,dz + \int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx \quad \text{—— (1)}$$

when $R \to \infty$, $\int_{C_R} \frac{z^2}{(z^2+1)(z^2+4)}\,dz$

$$= \int_{C_R} \frac{z^2}{z^2\left[1+\frac{1}{z^2}\right] z^2\left[1+\frac{4}{z^2}\right]}\,dz$$

$$\Longrightarrow 0$$

① $\Rightarrow$

$$\int_C \frac{z^2}{(z^2+1)(z^2+4)}\,dz = \int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx.$$

Using Residue Theorem

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx = \int_C \frac{z^2}{(z^2+1)(z^2+4)}\,dz$$

$$= 2\pi i \left[[\mathrm{Res}\, f(z)]_{z=i} + [\mathrm{Res}\, f(z)]_{z=2i} \right] \quad ②$$

$z = +i, 2i$ are poles inside C

$$[\mathrm{Res}\, f(z)]_{z=i} = \lim_{z\to i}\left[(z-i)\frac{z^2}{(z^2+1)(z^2+4)}\right]$$

$$= \lim_{z\to i}\left[\frac{z^2}{(z+i)(z^2+4)}\right]$$

$$= \frac{-1}{2i(-1+4)} = \frac{-1}{6i}$$

$$[\mathrm{Res}\, f(z)]_{z=2i} = \lim_{z\to 2i}\left[(z-2i)\frac{z^2}{(z^2+1)(z^2+4)}\right]$$

$[z^2+4 = (z+2i)(z-2i)]$

$$= \lim_{z\to 2i}\left[\frac{z^2}{(z^2+1)(z+2i)}\right]$$

$$= \frac{(2i)^2}{[(2i)^2+1]\,4i} = \frac{-4}{-3\times 4i} = \frac{1}{3i}$$

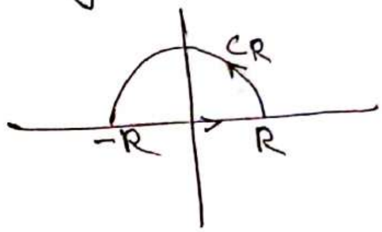
Substituting in ②

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)} dx$$

$$= 2\pi i \left[\frac{-1}{6i} + \frac{1}{3i} \right]$$

$$= 2\pi \left[\frac{-1}{6} + \frac{1}{3} \right] = \frac{\pi}{3}$$

(2) Evaluate $\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx$, using contour integration.



$$f(z) = \frac{z^2 - z + 2}{z^4 + 10z^2 + 9}.$$

$$= \frac{z^2 - z + 2}{(z^2+9)(z^2+1)}$$

$z = \pm i, \pm 3i$ are the poles.

Consider $\int_C f(z) dz$ where C is the upper half of the circle $|z| = R$ and the real axis from $-R$ to R.

$$\int_C f(z) dz = \int_{C_R} f(z) dz + \int_{-R}^{R} f(x) dx.$$

when $R \to \infty$,

$$\int_C f(z) dz = \int_{C_R} f(z) dz + \int_{-\infty}^{\infty} f(x) dx.$$

$$\int_C \frac{z^2 - z + 2}{(z^2+9)(z^2+1)} dz = \int_{C_R} \frac{z^2 - z + 2}{(z^2+9)(z^2+1)} dz + \int_{-\infty}^{\infty} \frac{x^2 - x + 2}{(x^2+9)(x^2+1)} dx. \quad \text{(1)}$$

when $R \to \infty$

$$\int_{C_R} \frac{z^2 - z + 2}{(z^2+9)(z^2+1)}\, dz = \int_{C_R} \frac{z^2\left[1 - \frac{1}{z} + \frac{2}{z^2}\right]}{z^2\left[1+\frac{9}{z^2}\right] z^2\left[1+\frac{1}{z^2}\right]}\, dz$$

$$\Longrightarrow 0 \quad \text{when } z \to \infty.$$

$$① \Rightarrow \int_{-\infty}^{\infty} \frac{x^2 - x + 2}{(x^2+9)(x^2+1)}\, dx = \int_{C} \frac{z^2 - z + 2}{(z^2+9)(z^2+1)}\, dz$$

$$\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9}\, dx = 2\pi i \left[\text{Res } f(z)\Big]_{z=3i} + \text{Res } f(z)\Big]_{z=i} \right] \quad ②$$

$\left[\begin{array}{l} z = 3i \text{ and } i \text{ lies inside } C \\ z = -3i \text{ and } -i \text{ lies outside } C \end{array}\right]$, Simple poles.

$$\text{Res } f(z)\Big]_{z=3i} = \lim_{z\to 3i} \left[\frac{(z-3i)(z^2 - z + 2)}{(z^2+9)(z^2+1)} \right]$$

$\left\{ z^2 + 9 = (z+3i)(z-3i) \right\}$

$$= \lim_{z\to 3i} \frac{z^2 - z + 2}{(z+3i)(z^2+1)}$$

$$= \frac{-9 - 3i + 2}{(6i)(-9+1)} = \frac{-7-3i}{-48i} = \frac{7+3i}{48i}$$

$$\text{Res } f(z)\Big]_{z=i} = \lim_{z\to i} (z-i) \left[\frac{z^2 - z + 2}{(z^2+9)(z^2+1)} \right]$$

$$= \lim_{z\to i} \frac{z^2 - z + 2}{(z^2+9)(z+i)}$$

$\left\{ \text{Cancel } z-i \right\}$

$$= \frac{-1-i+2}{(-1+9)(2i)} = \frac{1-i}{16i}$$

Substituting residues in (2)

$$\int_{-\infty}^{\infty} \frac{x^2-x+2}{(x^2+9)(x^2+1)}\,dx = 2\pi i\left[\frac{7+3i}{48i} + \frac{1-i}{16i}\right]$$

$$= 2\pi\left[\frac{7+3i}{48} + \frac{1-i}{16}\right]$$

$$= \frac{5\pi}{12}$$

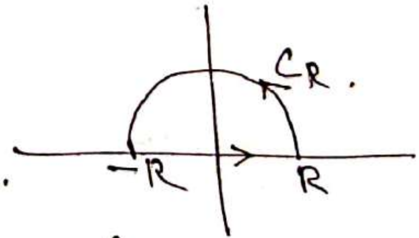
(3) Evaluate $\int_0^\infty \frac{\cos ax}{x^2+1}\,dx$ using Residue Theorem.

$$\int_0^\infty \frac{\cos ax}{x^2+1} = \text{Real part of } \int_0^\infty \frac{e^{iax}\,dx}{x^2+1}.$$

$$\boxed{e^{iax} = \cos ax + i\sin ax.}$$

$$f(z) = \frac{e^{iaz}}{z^2+1}.$$

$\int_C f(z)\,dz$, when C is the upperhalf of the circle $|z| = R$ and the real axis from $-R$ to R



$$\int_C f(z)\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx.$$

$$\int_C \frac{e^{iaz}}{z^2+1}\,dz = \int_{C_R} \frac{e^{iaz}}{z^2+1}\,dz + \int_{-R}^{R} \frac{e^{iax}}{(x^2+1)}\,dx$$

when $R \to \infty$

$$\int_C \frac{e^{iaz}}{z^2+1}\,dz = \int_{C_R}\left(\frac{e^{iaz}}{z^2+1}\right)dz + \int_{-\infty}^{\infty}\frac{e^{iax}}{x^2+1}\,dx.$$

when $\to \infty$ $\displaystyle\int_{C_R}\frac{e^{iaz}}{z^2+1}\,dz \Rightarrow 0$.

$$\therefore \int_C \frac{e^{iaz}}{z^2+1}\,dz = \int_{-\infty}^{\infty}\frac{e^{iax}}{x^2+1}\,dx.$$

$z = \pm i$ are simple poles.

$z = i$ lies inside C

$z = -i$ lies outside C.

$$\therefore \int_{-\infty}^{\infty}\left(\frac{e^{iax}}{x^2+1}\right)dx = \int_C \frac{e^{iaz}}{z^2+1}\,dz.$$

$$= 2\pi i\left[\operatorname{Res} f(z)\right]_{z=i}$$

$$= 2\pi i\left[\lim_{z\to i}(z-i)\frac{e^{iaz}}{z^2+1}\right]$$

$$= 2\pi i\left[\lim_{z\to i}\frac{e^{iaz}}{z+i}\right]$$

$$\int_{-\infty}^{\infty}\frac{e^{iax}}{x^2+1}\cancel{dx} = 2\pi i\left[\frac{e^{-a}}{2i}\right] = \pi e^{-a}$$

$$2\times\int_0^{\infty}\frac{e^{iax}}{x^2+1}\,dx = \pi e^{-a}.$$

$$\therefore \int_0^{\infty}\frac{e^{iax}}{x^2+1}\,dx = \frac{\pi}{2}e^{-a}$$

$$\int_0^\infty \left(\frac{\cos ax + i \sin ax}{x^2+1}\right) dx = \frac{\pi}{2} e^{-a}.$$

Equating Real parts $\displaystyle\int_0^\infty \frac{\cos ax}{x^2+1} dx = \frac{\pi}{2} e^{-a}.$

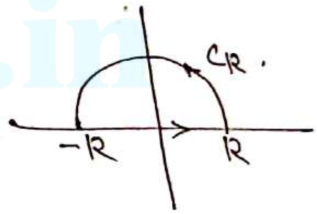
(4) Show that $\displaystyle\int_0^\infty \frac{\sin ax}{x^2+1} dx = 0.$

Same answer of question (3), Last taking imaginary parts.

$$\int_0^\infty \frac{\sin ax}{x^2+1} dx = 0.$$

(5) Using Contour integration, evaluate

$$\int_0^\infty \frac{1}{(x^2+a^2)^2} dx.$$



$$f(z) = \frac{1}{(z^2+a^2)^2}.$$

$\displaystyle\int_C f(z)\,dz$, where C is a closed contour consisting of upper half of the circle $|z|=R$ and real axis from $-R$ to R.

$$\int_C f(z)\,dz = \int_{C_R} f(z)\,dz + \int_{-R}^{R} f(x)\,dx.$$

ie $$\int_C \frac{1}{(z^2+a^2)^2} dz = \int_{C_R} \frac{1}{(z^2+a^2)^2} dz + \int_{-R}^{R} \frac{1}{(x^2+a^2)^2} dx.$$

when $R \to \infty$.

$$\int_C \frac{1}{(z^2+a^2)^2} dz = \int_{C_R} \frac{1}{(z^2+a^2)^2} dz + \int_{-\infty}^{\infty} \frac{1}{(x^2+a^2)^2} dx \quad \text{—— (1)}$$

when $R \to \infty$ $\displaystyle\int_{C_R} \frac{1}{(z^2+a^2)^2} dz$

$$= \int_{C_R} \frac{1}{z^4\left[1+\frac{a^2}{z^2}\right]^2} dz$$

$$\Longrightarrow 0$$

$\therefore$ (1) $\Longrightarrow$

$$\int_{-\infty}^{\infty} \frac{1}{(x^2+a^2)^2} dx = \int_C \frac{1}{(z^2+a^2)^2} dz$$

$z = \pm ai$ are poles of order 2.

$z = ai$ lies inside C

$z = -ai$ lies outside C.

$$= 2\pi i \left[\operatorname{Res} f(z) \right]_{z=ai}$$

$$= 2\pi i \left[\lim_{z \to ai} \frac{1}{(2-1)!} \frac{d}{dz} (z-ai)^2 \frac{1}{(z^2+a^2)^2} \right]$$

$$= 2\pi i \lim_{z \to ai} \frac{d}{dz} (z-ai)^2 \frac{1}{(z-ai)^2 (z+ai)^2}$$

$$= 2\pi i \lim_{z \to ai} \frac{d}{dz} \left[\frac{1}{(z+ai)^2}\right]$$

$$= 2\pi i \left[\lim_{z \to ai} \frac{-2}{(z+ai)^3} \right]$$

$$= 2\pi i \left[\frac{-2}{(2ai)^3} \right]$$

$[i^3 = -i]$

$$\int_{-\infty}^{\infty} \frac{1}{(x^2+a^2)^2} dx = \frac{-4\pi i}{8a^3 i^3} = \frac{\pi}{2a^3}$$

$$\therefore 2 \times \int_0^{\infty} \frac{1}{(x^2+a^2)^2} dx = \frac{\pi}{2a^3}$$

$$\therefore \int_0^{\infty} \frac{1}{(x^2+a^2)^2} dx = \frac{\pi}{4a^3}$$

H.W

(1) Evaluate $\int_{-\infty}^{\infty} \frac{x}{(x^2+1)(x^2+2x+2)} dx$, using residue theorem.

(2) Evaluate $\int_0^{\infty} \frac{x \sin x}{x^2+a^2} dx$.

$\left[= \text{Ima. part of } \int_0^{\infty} \frac{x e^{ix}}{x^2+a^2} dx \right]$